



هيئة المساحة الجيولوجية السعودية
Saudi Geological Survey

Pre-Qualification Response

Geoscience Data Analytics Center
Saudi Arabia's Advanced Analytics Platform

RFQ Reference Number:	251140007625
Government Entity:	Saudi Geological Survey (SGS)
Applicant:	Lithodat Pty Ltd
ABN:	63 627 008 904
Submission Deadline:	24/12/2025
Submission Platform:	Etimad Portal

Document Version: 1.0 | Date: December 2025
Prepared by: Keith Dimech, Chief Operating Officer

Document Control

Item	Details
Document Title	GDAC-SA Pre-Qualification Response
Version	1.0
Date Prepared	2025-12-09
Prepared By	Keith Dimech, Chief Operating Officer
Authorized Signatory	Dr. Fabian Kohlmann, Managing Director
Company	Lithodat Pty Ltd
ABN	63 627 008 904
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Currency Note: All financial figures are presented in both Australian Dollars (AUD) and Saudi Riyals (SAR) at exchange rate **AUD 1 = SAR 2.45**

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1. Executive Summary

Lithodat Pty Ltd is an Australian geoscience technology company specializing in FAIR-compliant geochemistry data platforms. We have developed and operate national-scale geoscience data infrastructure for the Australian government (EarthBank/AusGeoChem) and have delivered similar platforms for Natural Resources Canada.

7+

YEARS EXPERIENCE

\$3.7M

AUD PROJECT VALUE

SAR 9,065,000

19

TEAM MEMBERS

Key Qualifications Summary

Criterion	Lithodat Capability
Experience	7+ years developing national geoscience data platforms
Similar Projects	4 major projects: AUD \$3,700,000 / SAR 9,065,000
Technical Team	19 staff including 5 PhDs in geoscience and data science
Government Experience	NCRIS (Australia), NRCan (Canada), CSIRO, Geoscience Australia
Financial Health	Cash Ratio 2.05, Current Ratio 3.21, Quick Ratio 3.19
AI/ML Capability	Dr. Behnam Sadeghi (IAMG Vistelius Award winner, 58+ publications)
Saudi Expertise	Dr. Mahdi AbuAli (35+ years Saudi Aramco), Dr. Qusay Abeed (Native Arabic)

Competitive Advantage: Lithodat is the only applicant with proven national-scale geoscience data platform delivery (EarthBank: 350,000+ samples, 1,300+ users), combined with dedicated Saudi Arabian expertise through Dr. Mahdi AbuAli's 35+ years at Saudi Aramco.

2. Form 9.1 - Applicant Information

Company Details

Field	Response
Name of the applicant (company/institution)	Lithodat Pty Ltd
Value of the applicant's capital	AUD \$370,575 SAR 907,909
Date Function	Active
Commercial Registration Number	ABN 63 627 008 904 (Australia)
Name of the person responsible	Dr. Fabian Kohlmann

Address

Field	Response
Street	24 Crompton Way
City	Newtown
Zip Code	3220
Country	Australia
State/Province	Victoria

Contact Information

Field	Response
Phone	+61 3 5215 5129
Email	info@lithodat.com
Website	https://www.lithodat.com
Establishment Date	25 June 2018
Years in Operation	7 years

Company Ownership Structure

#	Owner/Partner Name	Partnership %	Nationality
1	Dr. Fabian Kohlmann	50.1%	German/Australian
2	Dr. Wayne Noble	14.9%	Australian
3	Gerd Moritz Theile	10.0%	German
4	Vinko Novak	10.0%	Croatian
5	Enreach Resources Pty Ltd	15.0%	Australian
Total		100%	

Company Representative (Contact Administrator)

Field	Response
Name	Keith Dimech
Job Title	Chief Operating Officer / Project Director
Phone	+61 3 5215 5129
Mobile	+61 428 971 396
Email	keith.dimech@lithodat.com
Alternate Contact	Dr. Fabian Kohlmann (fabian.kohlmann@lithodat.com)

3. Form 9.2 - Technical and Administrative Capabilities

A. Previous Work Experience

Question	Response
Number of years of experience in the field of project	7 years (2018-Present)
Number of similar projects executed in the last three years	4 projects
Average results of performance evaluations	Excellent

Similar Projects Summary (Last 3 Years)

#	Project	Client Type	Value (AUD)	Value (SAR)	Status
1	EarthBank (AusGeoChem)	NCRIS/Government	\$2,900,000	7,105,000	Ongoing
2	Isotopes.au	CSIRO/Government	\$400,000	980,000	Ongoing
3	NRCan ThermoChronology	Federal Govt (Canada)	\$400,000	980,000	DELIVERED
4	LithoSpace	Internal R&D	Internal	–	Ongoing
TOTAL			\$3,700,000	9,065,000	

B. Quality Assurance

Standard/Framework	Status	Application
CoreTrustSeal	Certified	Data repository certification (EarthBank)
ISO 9001:2015	Principles Applied	Quality management system
FAIR Data Principles	Implemented	Findable, Accessible, Interoperable, Reusable
Peer Review Process	Active	Scientific validation
Agile/Scrum	Implemented	Development methodology

C. Environment, Health and Safety

Standard/Framework	Status	Application
ISO 14001:2015	Principles Applied	Environmental management
ISO 45001:2018	Principles Applied	Occupational health and safety
Remote Work Safety	Implemented	Distributed team protocols
Data Security	Expert	Vinko Novak - Head of Data Security

HSE Experience: Keith Dimech (COO) brings 15 years managing Environmental, Safety, and Quality systems in high-risk construction including: \$5.1B Snowy Hydro 2.0, \$11B Melbourne Metro Tunnel, and \$900M Melbourne Water projects.

D. Existing Contractual Obligations

Question	Response
Number of existing projects	3
Value of existing projects (AUD)	\$3,600,000
Value of existing projects (SAR)	8,820,000

Current Projects Detail

#	Project	Client	Value (AUD)	Value (SAR)	End Date
1	EarthBank Platform	AuScope/NCRIS	\$2,900,000	7,105,000	Ongoing
2	National Thermochronology	NRCan	\$300,000	735,000	2026
3	Isotopes.au Platform	CSIRO/ANSTO/GA	\$400,000	980,000	2028
TOTAL			\$3,600,000	8,820,000	

E. Human Resources

Question	Response
Number of employees	19
Number of Saudi employees	0
Percentage of Saudi employees	0.00%

Workforce Composition

Category	Count	Details
Management/Directors	3	Fabian Kohlmann (CEO), Wayne Noble (CIO), Keith Dimech (COO)
Full-Time Technical Staff	10	4 developers, 1 lead developer, 1 data security, 1 ops manager, 3 specialists
Contractors/Consultants	6	ML advisor, AI lead, geological consultant, data quality specialists
Total	19	

Saudization Commitment

Lithodat is committed to supporting Saudi Vision 2030 workforce development objectives:

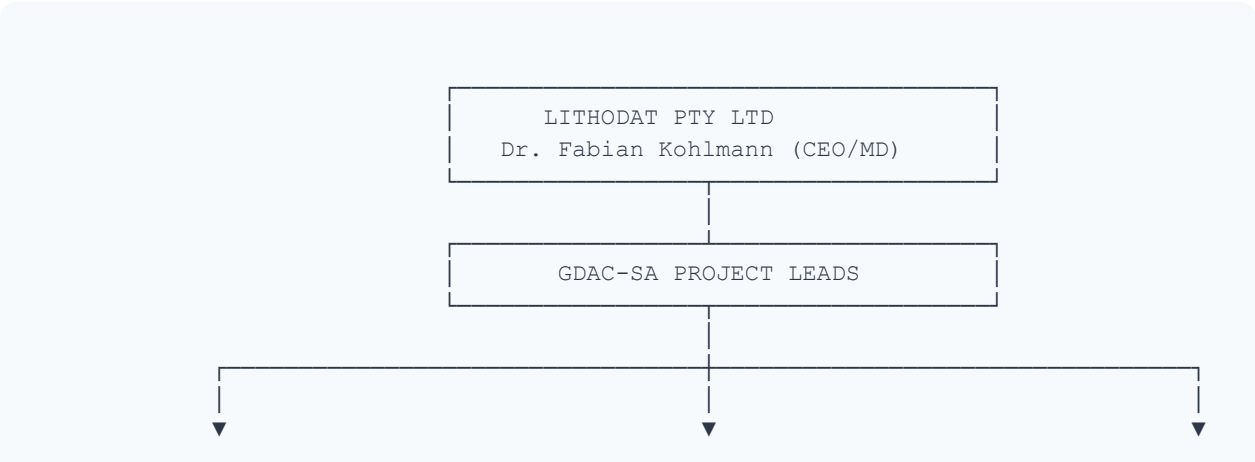
Phase	Timeline	Saudi Hires	Roles
Phase 1	Year 1	2	Junior developers, data analysts
Phase 2	Year 2	3	Mid-level technical, project support
Phase 3	Year 3+	5+	Senior technical, management roles

4. Form 9.3 - Administrative Staff Experience

RFQ Requirement: "Describe the experiences of the applicant's administrative staff as follows:" (10 rows)

#	Name	Function	Specialization / Area of Expertise	Years Exp.
1	Dr. Fabian Kohlmann	Managing Director & CEO	Geoscience Leadership, FAIR Data Architecture, Thermochronology, Strategic Planning	15+ years
2	Dr. Wayne Noble	Chief Information Officer	Enterprise Architecture, Cloud Systems, Agile Leadership, Software Development	25+ years
3	Keith Dimech	Chief Operating Officer	Project Management, ESQ Systems, Financial Operations, HR, Legal, HSEQ	15+ years
4	Gerd Moritz Theile	Lead Developer / Director	PostgreSQL Architecture, Database Design, LithoSurfer Platform, Java	25+ years
5	Juan Baca	Operations Manager (LATAM)	Data Quality Management, Team Leadership, Database Operations, Training	10+ years
6	-	-	-	-
7	-	-	-	-
8	-	-	-	-
9	-	-	-	-
10	-	-	-	-

GDAC-SA Project Leadership Structure



IN-KINGDOM DIRECTOR Dr. Mahdi AbuAli Government Relations & Client Value Alignment	TECHNICAL DIRECTOR Dr. Qusay Abeed Technical Implementation & Geological Data Systems	PROJECT DIRECTOR Keith Dimech Project Management & Operations
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Key Administrative Staff Profiles

Dr. Fabian Kohlmann - Managing Director & CEO

Field	Details
Position	Managing Director & Chief Geoscientist
Shareholding	50.1%
Years with Lithodat	7 years (Founder, December 2017)
Total Experience	15+ years
Education	PhD, University of Melbourne (2010)
Key Competencies	FAIR geochemistry frameworks, relational database architecture, SKOS vocabularies, thermal history modeling

Dr. Wayne Noble - Chief Information Officer

Field	Details
Position	Chief Information Officer / Executive Director
Shareholding	14.9%
Years with Lithodat	6+ years (Since December 2018)
Total Experience	25+ years
Education	PhD Geology, La Trobe University (1998); Graduate Diploma Computer Science (with Distinction), RMIT (2006); BSc (Hons) Geology First Class, La Trobe (1991)
Key Competencies	Java, Python, JavaScript, cloud-native applications (AWS), microservices, API design, Agile/Scrum

Keith Dimech - Chief Operating Officer

Field	Details
Position	Chief Operating Officer
Years with Lithodat	May 2025 - Present (7 months)
Total Experience	15+ years
Education	Executive MBA (RMIT, 2025); Bachelor (Hons) Geoscience (University of Melbourne, 2014)
Key Competencies	Project portfolio management, HR, budgeting, ESQ (15 years), SQL databases, ISO 9001/14001/45001

Lithodat COO Achievements (May 2025 - Present):

- Executive leadership of \$4.2M+ active project portfolio
- Recruited 4 new technical staff members
- Restructured company financial systems
- Executed 3 major client contracts (\$1.8M new revenue)
- Established OKR framework across all departments
- Led strategic planning for GDAC-SA tender submission

5. Form 9.4 - Professional Staff Experience

RFQ Requirement: "The expertise of the applicant's technical staff is as follows:" (20 rows)

#	Name	Function	Specialization / Area of Expertise	Years Exp.
1	Dr. Mahdi AbuAli	GDAC In-Kingdom Director	Saudi Government Relations, Saudi Aramco (35+ years), Executive Leadership, JV Program Management, Petroleum Systems	35+ years
2	Dr. Qusay Abeed	GDAC Technical Director	Petroleum Geochemistry, Basin Modelling, CO2 Storage (CCUS), Native Arabic Speaker, C++, Agile Scrum	17+ years
3	Dr. Behnam Sadeghi	ML Technical Advisor	ML/AI, Geospatial AI, Python (PyImpetus, PyMiner), IAMG Vistelius Award, Fulbright Fellow, CSIRO	15+ years
4	Pedro H. C. Ferreira	AI Software Dev Lead	Technology & AI/ML Strategy, Enterprise AI Implementation	10-15 years
5	Vinko Novak	Head of Data Security	Data Security, Information Protection, Compliance, Security Architecture	10+ years
6	Xinyan Zhang	Frontend Developer	React, TypeScript, UI/UX implementation, Responsive design	3+ years
7	Lujia Yang	Frontend Developer	React, TypeScript, UI/UX implementation, Responsive design	3+ years
8	Tarun Sengar	Backend Developer	Java, AWS, Backend Systems, API Development	5+ years
9	Nirali Dudharejiya	Backend Developer	Java, Liquibase, Backend Systems, Database Management	5+ years
10	Cris Ibarra	Data Quality Specialist	Data Entry, Data Validation, Quality Control, Database Operations	3+ years
11	Perla Luque	Data Quality Specialist	Data Entry, Data Validation, Quality Control, Database Operations	3+ years
12	Dr. Alejandra Bedoya	Data Entry / PhD Candidate	Geochronology, Thermochronology, Tectonics, Data Validation	10+ years
13	Dr. Fun Meeuws	Geochemistry Specialist	Fission Track Thermochronology, Thermal History Modelling, GPLates	15+ years

14	Benjamin Dib	Data Clerk (LithoSpace)	Space Science, Geospatial Data, Moon/Mars Mission Data, R Programming	3+ years
15- 20	Reserved for additional contractors as required			

Key Technical Personnel Profiles

Dr. Mahdi AbuAli - GDAC In-Kingdom Director

Field	Details
Current Position	GDAC In-Kingdom Director (Specialist Contractor)
Other Positions	COO & VP - Raphael Energy Group; MD - Fleet Tahreez Company (Riyadh)
Location	Saudi Arabia / Australia
Languages	English (Fluent), Arabic (Native - Saudi dialect)
Saudi Aramco Tenure	35+ years (1985-2020)

GDAC-SA Role: Government Relationship Management, Client Value Alignment, Kingdom Delivery Assurance

Key Achievements:

- 35+ years Saudi Aramco - Rising from Geologist to Senior Geological Consultant
- Managed >200,000 km² of JV exploration acreage with ENI, Shell, Lukoil
- Matured and risked 50+ prospects - several resulting in discoveries
- Recent Riyadh presence - MD role with satellite/AI exploration workflows (2025)

Dr. Qusay Abeed - GDAC Technical Director

Field	Details
Current Position	Senior Geological Consultant / Petroleum Geochemist
Location	Oxford, United Kingdom
Languages	English (Fluent), Arabic (Native)
Total Experience	17+ years
Education	PhD Petroleum Geochemistry, RWTH Aachen (magna cum laude)

Value for GDAC-SA:

- **Native Arabic Speaker** - Direct communication with SGS stakeholders
- **Middle East Geology Expertise** - Deep understanding of regional systems
- **CCUS Knowledge** - Aligns with Saudi Vision 2030 sustainability
- **Halliburton Connection** - Former colleague of Dr. Kohlmann

Dr. Behnam Sadeghi - Machine Learning Technical Advisor

Field	Details
Current Position	ML Technical Advisor (CSIRO Research Fellow)
Total Experience	15+ years
Education	PhD Mining Engineering (Data Analytics), UNSW (2020)

Recognition:

- **2023 IAMG Vistelius Research Award** - Highest honor for early-career mathematical geoscientists
- **Fulbright Fellow** - Stanford University (2022-2023)
- **58+ peer-reviewed publications** (H-index: 24, 2000+ citations)

Published Software:

Package	Description	GDAC-SA Application
PyImpetus	Feature selection	Identifying geochemical pathfinders
PyMiner	Prospectivity mapping	Exploration targeting
PySensibo	Time series analysis	Temporal trends

6. Form 9.5 - Similar Projects

RFQ Requirement: "Write details of three similar projects that have been implemented over the past five years"

Project 1: EarthBank (AusGeoChem)

Field	Response
1. Project Name	EarthBank (formerly AusGeoChem) - AuScope Geochemistry Network Platform
2. Project Location	Australia (National infrastructure, cloud-hosted)
3. Project Content	National FAIR data platform for geochemistry research. 350,000+ samples, 1,300+ users, custom SKOS schema with 50,000+ terms. DOI/IGSN minting, REST API, map visualization, LithoPlates paleogeographic reconstruction.
4. Project Owner	AuScope Geochemistry Network (AGN) / AuScope Limited (NCRIS)
5. Contract Value	AUD \$2,900,000 SAR 7,105,000
6. Contract Duration	5+ years (2020 - Present, ongoing)
7. Start Date	2020
8. Delivery Date	Platform live October 2021; Official launch late 2022
9. Name of person responsible	Professor Brent McInnes
10. Contact information	John de Laeter Research Centre, Curtin University, Perth, WA
11. Email	directorjdlc@curtin.edu.au

Evidence: Chemical Geology (2025) publication | CoreTrustSeal certified | <https://earthbank.auscope.org.au/>

Project 2: Isotopes.au

Field	Response
1. Project Name	Isotopes.au: National Data Infrastructure for Environmental Isotope Analysis
2. Project Location	Australia (National infrastructure, cloud-hosted)
3. Project Content	National platform harmonising isotope data from 6 institutional sources (CSIRO, ANSTO, GA, NMI). Unified schema with 250+ field mappings, 97% ontology match rate, automated adapters.
4. Project Owner	CSIRO / ANSTO / Geoscience Australia / NMI Consortium
5. Contract Value	AUD \$400,000 SAR 980,000
6. Contract Duration	4.5 years (July 2024 - December 2028)
7. Start Date	July 2024
8. Delivery Date	Development complete; Platform deployment in progress
9. Name of person responsible	Lian Flick
10. Contact information	CSIRO Data61, Australia
11. Email	Lian.Flick@data61.csiro.au

Evidence: R&D Registration IISA0053835 | Experiment Reports (Adapter, Wizard, Ontology Validation)

Project 3: NRCan National Thermochronology Platform

Field	Response
1. Project Name	NRCan National Thermochronology & Thermal History Data Model Development
2. Project Location	Canada (nationwide)
3. Project Content	National unified data model for thermochronology (AHe, ZHe, AFT, ZFT). Complete standardisation of all provincial and historic datasets. Automated ingestion and QA/QC pipelines. 100% Canadian legacy data harmonised.
4. Project Owner	Natural Resources Canada (Geological Survey of Canada)
5. Contract Value	AUD \$400,000 SAR 980,000
6. Contract Duration	2 years (October 2021 - March 2023)
7. Start Date	October 2021
8. Delivery Date	March 2023 (DELIVERED)
9. Name of person responsible	Dr. Jeremy Powell
10. Contact information	Natural Resources Canada, Ottawa
11. Email	jeremy.powell@nrcan-rncan.gc.ca
Status: DELIVERED Contract documentation and GSC sign-off available	

Projects Summary

Project	Client	Value (AUD)		Value (SAR)	Duration	Status
EarthBank	NCRIS/AuScope	\$2,900,000		7,105,000	2020-Present	Production
Isotopes.au	CSIRO/ANSTO/GA	\$400,000		980,000	2024-2028	Development
NRCan	Geological Survey of Canada	\$400,000		980,000	2021-2023	DELIVERED
TOTAL		\$3,700,000		9,065,000		

7. Form 9.6 - Financial Capacity Criteria

Currency Information: All figures presented in both Australian Dollars (AUD) and Saudi Riyals (SAR) at exchange rate **AUD 1 = SAR 2.45**

Balance Sheet Statement (Last 3 Years)

Line Item	FY2025 (Year 0)		FY2024 (Year -1)		FY2023 (Year -2)	
	AUD	SAR	AUD	SAR	AUD	SAR
Current Assets	\$430,819	1,055,507	\$176,800	433,160	\$93,690	229,541
Cash & Cash Equivalents	\$274,855	673,395	\$100,406	246,095	\$40,605	99,482
Accounts Receivable	\$152,964	374,762	\$76,394	187,166	\$53,001	129,852
Current Liabilities	\$134,211	328,817	\$40,106	98,260	\$36,787	90,128

Financial Performance Indicators

CASH RATIO

2.05

4x above 0.5 benchmark

CURRENT RATIO

3.21

2x above 1.5 benchmark

QUICK RATIO

3.19

3x above 1.0 benchmark

Indicator	Formula	FY2025	FY2024	FY2023	Benchmark	Assessment
Cash Ratio	Cash ÷ Current Liabilities	2.05	2.50	1.10	>0.5	Excellent
Turnover Ratio	Current Assets ÷ Current Liabilities	3.21	4.41	2.55	>1.5	Excellent
Quick Liquidity Ratio	(Cash + Receivables) ÷ Current Liabilities	3.19	4.41	2.54	>1.0	Excellent

Additional Financial Metrics

Metric	FY2025 (AUD)	FY2025 (SAR)	Assessment
Net Working Capital	\$296,608	726,690	Strong
Revenue (FY2025)	\$1,192,265	2,921,049	+76% YoY
Net Profit After Tax	\$159,914	391,789	+100% YoY
Revenue Growth (3-year CAGR)	62.8%		Strong Growth
Debt to Equity Ratio	0.45:1		Low Leverage

Financial Statements Attachments

#	Document	Fiscal Year	Status
1	Annual Report (Financial Statements)	FY2023	Available
2	Annual Report (Financial Statements)	FY2024	Available
3	Annual Report (Financial Statements)	FY2025	Available
4	Company Tax Return	FY2023	Available
5	Company Tax Return	FY2024	Available
6	Company Tax Return	FY2025	Available

8. Supporting Documentation Index

Required Documents Checklist

#	Document	RFQ Reference	Status
1	Valid Commercial Registration Certificate	Page 8	AVAILABLE
2	Valid Zakat and Income Certificate / VAT	Page 8	N/A - Australian Entity
3	General Organization for Social Insurance	Page 8	N/A - Australian Entity
4	Valid Chamber of Commerce Certificate	Page 8	AVAILABLE
5	Valid investment license (MISA)	Page 8	IN PROGRESS
6	Valid Saudization certificate	Page 8	N/A - Saudization Plan Provided
7	Quality policy/certificate and HSE policy	Page 8	AVAILABLE
8	Alliance agreement (if applicable)	Page 8	N/A - Single Entity

9	Approved financial statements (3 years)	Page 8	AVAILABLE
10	Company information and registration certificates	Page 8	AVAILABLE

9. Declaration

I hereby certify that all information provided in this qualification application is true, accurate, and complete to the best of my knowledge. The projects described were implemented by Lithodat Pty Ltd within the past five years as stated. All financial information is consistent with the attached audited financial statements.

Lithodat Pty Ltd undertakes to maintain the confidentiality of any information or data obtained during the qualification process and not to disclose any such information to any third party without written permission from the Saudi Geological Survey.

Authorized Signatory

Company Name:	Lithodat Pty Ltd
ABN:	63 627 008 904
Name:	Dr. Fabian Kohlmann
Title:	Managing Director

Signature

Date

Prepared By

Name:	Keith Dimech
Title:	Chief Operating Officer
Email:	keith.dimech@lithodat.com
Phone:	+61 428 971 396

Signature

Date

[COMPANY SEAL]



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Lithodat Pty Ltd
ABN 63 627 008 904
December 2025